

# What to Measure

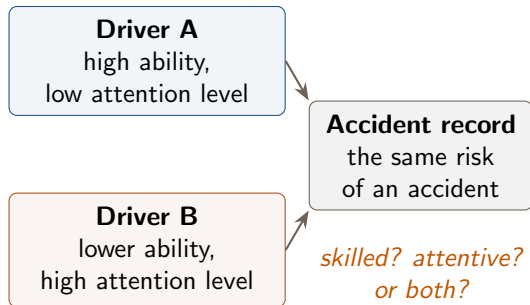
## Optimal Monitoring under Adverse Selection and Moral Hazard

Xinrong Liu

The University of Texas at Austin

Accident records mix driving ability and attention.  
Which additional measure improves contracts, behavior, and welfare?

# One record, two reasons to be safe



- The insurer observes the **accident record**.
- Safety can come from **ability** (a private characteristic) or **attention level** (a hidden choice).
- The insurer uses contracts to **screen ability** and **encourage attention**.
- The same record can come from different combinations of ability and attention.

# The questions

**The problem.** The accident record mixes ability and attention level.

*the insurer can add one signal*

**1. What should a new telematics signal measure:** ability, attention level, or a mix?

*information serves two economic roles*

**2. Which signal direction is more valuable:** one that better solves the incentive problem, or one that better resolves the information problem?

*the signal changes the optimal contract*

**3. How does better monitoring change the allocation?** Does it raise attention levels across the board, or reallocate attention across drivers?

*allocation changes determine welfare*

**4. Who gains from better information,** and when does more monitoring raise total surplus?

# The plan

## 1. The record-only benchmark

How a bonus encourages attention when ability is private.

## 2. Find the optimal weak signal

Choose the signal direction and payments together. Then examine attention and welfare.

## 3. Why the weak-signal calculation works

Four steps connect the benchmark to a general direction criterion. The detailed argument follows the takeaways.

### The starting point

A weak signal changes the existing incentive problem only a little. Its value depends on which incentive requirements are costly at the benchmark.

# Where this sits

- **Adverse selection + moral hazard:** Laffont–Tirole (1986, 1993), Picard (1987), Guesnerie–Picard–Rey (1989), Myerson (1982); Castro-Pires–Chade–Swinkels (2024), Gottlieb–Moreira (2022). *They take the information structure as given.*
- **Value and design of information in agency:** Holmström (1979), Kim (1995), Holmström–Milgrom (1991), Lewis–Sappington (1997), Georgiadis–Szentes (2020). *There a signal is about the action; here one signal serves two frictions.*
- **Optimal experimental design:** determinant-optimal designs and Fisher information (Kiefer 1959; Pukelsheim 1993). *The statistical criterion the economic one is compared with.*
- **First-order approach / local IC:** Mirrlees (1999), Rogerson (1985), Jewitt (1988), Rochet–Choné (1998). *Does the relaxation stay valid as information moves?*
- **Telematics and usage-based insurance:** Reimers–Shiller (2019), Soleymanian–Weinberg–Zhu (2019), Jin–Vasserman (2021). *Monitoring changes behavior and selects safer drivers; which data to collect is a design choice.*

# Results: the general criterion and the example

## General result, under the stated regularity conditions

A weak signal is valuable when it relaxes costly incentive requirements. The best direction depends on the existing information, screening, and action incentives. **It can measure ability, attention, or a mix.**

### What the insurance calibration shows

#### Lessons from the insurance example

- **Incentive needs shape what to measure.** Targeting costly incentive constraints can move the optimal direction away from the statistically optimal one.
- **Signal direction helps explain how contracts change.** With weak monitoring, rewards tied to the attention signal strengthen incentives for less able drivers and weaken them for some abler drivers.
- **Private gains need not translate into social gains.** The insurer gains and drivers lose in this example. Total surplus can rise or fall, depending on changes in accident losses, effort costs, and payment risk.

# Benchmark: insurance without telematics

## The driver

- Private **ability**  $\theta \in [0, 1]$ . Higher  $\theta$  means greater driving ability.
- Hidden **attention level**  $a \in [0, 1]$ , at cost  $c(\theta, a)$ , increasing and convex in  $a$ , with  $c(\theta, 0) = 0$ .
- Risk-averse utility  $u(t) - c(\theta, a)$  from a payment  $t$ . Outside option: 0.

## The insurer

- Observes only the **accident record**  $y \in \{0, 1\}$ , where  $y = 1$  means an accident.
- Bears a loss  $\ell$  per accident and minimizes expected payments plus accident losses.
- Can offer transfers  $t \geq 0$ . Ability and attention are not contractible.

## The benchmark problem

Choose a menu of payments based on  $y$ , subject to truthful reporting and the driver's optimal attention choice. We first solve this problem **without the additional signal  $s$** .

# The accident record mixes ability and attention

Both greater ability and greater attention reduce accident risk:

$$p(\theta, a) = \Pr(y = 1 \mid \theta, a) = \Lambda(\gamma_0 - \gamma_\theta \theta - \gamma_a a), \quad \Lambda(z) = \frac{1}{1 + e^{-z}}.$$



- The record depends on  $(\theta, a)$  only through  $\gamma_\theta \theta + \gamma_a a$ .
- Driver A's greater ability offsets his lower attention. A and B therefore have the **same distribution of  $y$** .
- A reward for avoiding an accident rewards both sources of safety. The insurer cannot tie it to attention alone.

The illustration uses  $\gamma_\theta = 1.5$  and  $\gamma_a = 4$ . Gray lines join pairs with the same accident risk.

# Payments and the driver's attention

Write each transfer in **utility units**:  $v = u(t)$ . Its money cost is  $h(v) = u^{-1}(v)$ , which is convex under risk aversion.

| outcome                 | utility payment | money transfer  |
|-------------------------|-----------------|-----------------|
| accident ( $y = 1$ )    | $b$             | $h(b)$          |
| no accident ( $y = 0$ ) | $b + \Delta$    | $h(b + \Delta)$ |

Facing  $(b, b + \Delta)$  after (accident, no accident), a driver of ability  $\theta$  chooses  $a \in [0, 1]$  to maximize

$$b + (1 - p(\theta, a))\Delta - c(\theta, a).$$

- **The bonus changes attention.** At an interior choice,  $-p_a(\theta, a)\Delta = c_a(\theta, a)$ .
- **The base changes utility.** Raising both utility payments equally leaves the attention choice unchanged.

# The menu and all incentive constraints

A report  $\hat{\theta}$  selects utility payments  $(v_0(\hat{\theta}), v_1(\hat{\theta}))$  and *deterministic* recommended attention  $a(\hat{\theta})$ . The driver chooses **both the report and actual attention**.

$$\min_{a, v_0, v_1} \mathbb{E}_{\theta} [(1 - p(\theta, a(\theta)))h(v_0(\theta)) + p(\theta, a(\theta))h(v_1(\theta)) + \ell p(\theta, a(\theta))] .$$

Full incentive compatibility requires

$$U(\theta) \geq (1 - p(\theta, \hat{a}))v_0(\hat{\theta}) + p(\theta, \hat{a})v_1(\hat{\theta}) - c(\theta, \hat{a}) \quad \text{for every } (\hat{\theta}, \hat{a}).$$

- **Obedience (on-path):** report truthfully, change attention.
- **Screening:** change the report, follow that report's recommendation.
- **Joint deviations (obedience off-path):** take another contract and choose attention for one's true ability.

Under full IC,  $U(\theta)$  equals the driver's best attainable utility over *all* reports and attention choices, including off-path choices. Participation:  $U(\theta) \geq 0$ . Limited liability:  $v_0, v_1 \geq 0$ .

# A common bonus with interior attention

## Choosing the accident loss

- A larger loss makes preventing accidents more valuable.
- We choose  $\ell = 24$ : incentives matter and attention remains **interior**.
- Numerical benchmark: zero after an accident,  $\Delta^0 = 0.76$  utility units otherwise.

## Why a common contract can suffice

- At a fixed base, every driver prefers a larger bonus.
- At fixed utility, a larger bonus and a lower base induce more attention.
- If avoided losses exceed extra payment and risk costs *throughout the comparisons*, a common larger bonus lowers cost.

## The benchmark used below

Each driver chooses  $a^0(\theta)$  between 0.08 and 0.45. Here higher ability raises attention's marginal cost and lowers its benefit for accident prevention.

A uniform cost comparison is sufficient for a common contract (backup). The numerical result and a local check alone do not verify that condition.

# Why the record-only bonus is costly

## Pay needed for attention

$$\underbrace{\gamma_a p (1 - p) \Delta^0}_{\text{marginal incentive}} = \underbrace{c_a}_{\text{marginal attention cost}}.$$

Thus the expected utility payment is

$$\mathbb{E}[v] = (1 - p) \Delta^0 = \frac{c_a}{\gamma_a p}.$$

**When  $p$  is small**, each unit of attention incentive requires a large expected utility payment.

## The economic reason

1. **Most drivers receive the bonus.**  
Here 89% avoid an accident.
2. **Attention changes receipt only a little.**  
Avoiding an accident is already likely.
3. **Pay also rewards existing safety.**  
Able drivers collect the bonus even at low attention. Risk aversion adds a cost of uncertain pay.

## What an additional measure could do

A measure sensitive to attention could supply incentives with lower payments. Its value also depends on screening and payment risk.

# A numerical illustration

## Specification

- $\theta \sim U[0, 1]$ ,  $a \in [0, 1]$ .
- Attention cost:  $c(\theta, a) = \frac{1}{2}(0.75 + \theta)^2 a^2$ .
- Accident risk:  
 $p(\theta, a) = \Lambda(-0.5 - 1.5\theta - 4a)$ .
- Risk aversion:  $u(t) = (1 - e^{-\rho t})/\rho$ , so  
 $h(v) = -\ln(1 - \rho v)/\rho$ .

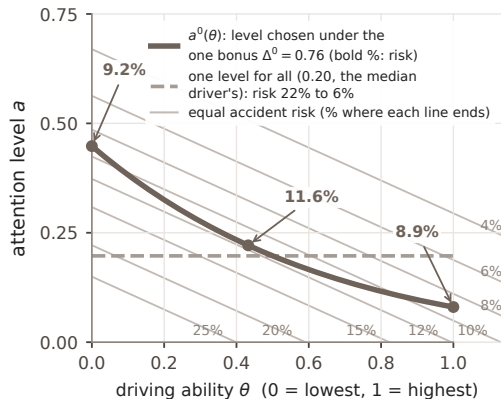
| $\rho$ | $\ell$ | $\gamma_0$ | $\gamma_\theta$ | $\gamma_a$ |
|--------|--------|------------|-----------------|------------|
| 0.5    | 24     | -0.5       | 1.5             | 4          |

## Economic meaning

- Ability and attention substitute in safety: one unit of ability offsets 0.375 units of attention in accident risk.
- Attention is costly for every driver, and more costly at the margin for drivers with greater ability.
- The loss  $\ell = 24$  makes accident prevention valuable. The optimal record-only attention choices remain strictly inside  $[0, 1]$ .

Numerical search enforces every report–attention deviation for the sampled types. We use 21 types, checked with 11 and 41. Payments are optimal at the computed attention levels. Allocation search is local (details in the backup).

# The benchmark contract and attention



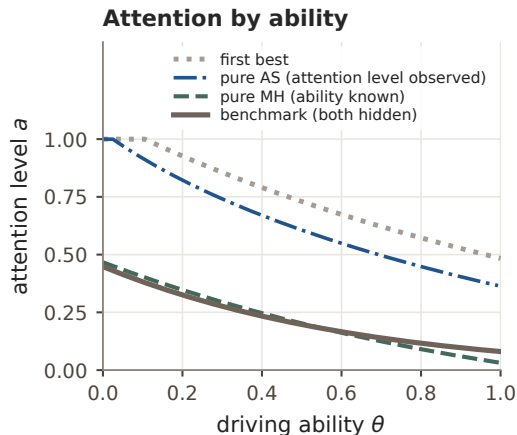
## One contract in the numerical solution

- No accident:  $\Delta^0 = 0.76$  utilities, or 0.95 in money. Accident: zero payment.
- Each driver chooses  $a^0(\theta)$ . Attention falls from 0.45 to 0.08 as ability rises.

## How the choices affect accident risk

- Lower attention partly offsets greater ability. Accident risks lie between 9% and 12%.
- Gray lines have equal risk. Drivers on one line generate the same distribution of accident records.

# Four reference allocations



## What each reference allows

- **First best:** ability known and attention contractible.
- **Pure AS:** attention contractible, ability private.
- **Pure MH:** ability known, attention hidden. The insurer can tailor the bonus to ability.
- **Benchmark:** both hidden. One common bonus in this calibration.

In this calibration, relative to **ability known**, the common bonus induces less attention at low ability and slightly more at high ability. New information may therefore change **how attention is allocated across drivers**.

# What another measure could change

## Insurer cost in this calibration

| reference  | payments plus losses |
|------------|----------------------|
| first best | 0.79                 |
| pure AS    | 1.09                 |
| pure MH    | 3.37                 |
| benchmark  | 3.41                 |

Money per driver. The comparisons make the incentive cost of hidden attention concrete.

## Two possible effects on contracts

- **Rewarding attention:** support a given attention level with less costly payments.
- **Sorting drivers:** relax reporting constraints and allow different incentives across abilities.

The record measures  $1.5\theta + 4a$ . The complementary measure  $1.5a - 4\theta$  separates the two sources of safety. We will compare its statistical value with its incentive value.

## The information-design question

Which measure most reduces insurer cost, once payments and attention choices can adjust?

# Signal design: precision and direction

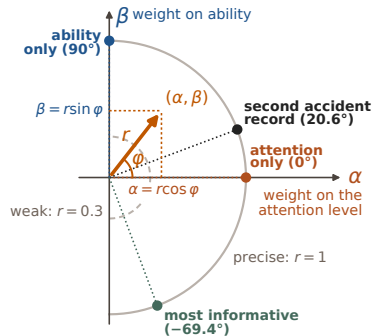
The signal is **pass or fail**. Conditional on  $(\theta, a)$ , it is independent of the accident record:

$$q(\theta, a) = \Pr(\text{pass} \mid \theta, a) = \Lambda(\alpha(a - a_c) + \beta(\theta - \theta_c)).$$

Write  $(\alpha, \beta) = r(\cos \varphi, \sin \varphi)$ .

- **Precision**  $r$ : how strongly the pass rate responds to the measured combination.
- **Direction**  $\varphi$ : the mix of attention and ability.  $0^\circ$  measures attention only,  $90^\circ$  ability only.
- A negative ability weight makes a more able driver less likely to pass at the same attention level.

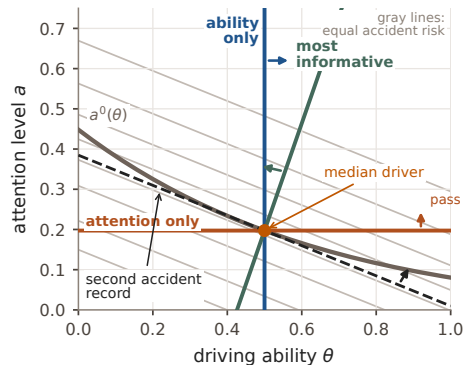
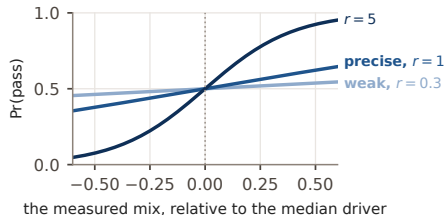
The test is centered at  $(\theta_c, a_c) = (0.5, 0.20)$ . Reversing both weights exchanges pass and fail.



Here distance from the origin is precision. We compare directions at the same  $r$ . A monitoring budget can allow any strength up to  $r$ .

# Four directions at the same precision

Greater precision makes the pass probability respond more sharply to the chosen mix.



Each line is a test threshold. Moving in the arrow's direction increases the pass probability.

D-optimal means maximizing the determinant of Fisher information about  $(a, \theta)$  at the benchmark, for a weak signal. The figure calls this direction “most informative.”

# Choosing a signal direction

## The economic calculation

1. Fix precision  $r$  and a candidate direction  $\varphi$ .
2. Redesign the payments and recommended attention, imposing **all** report–attention constraints.
3. Compare the lowest computed insurer cost with the record-only benchmark.

$$G(r, \varphi) = \text{benchmark cost} - \text{cost with the signal}, \quad g(\varphi) = \lim_{r \rightarrow 0} \frac{G(r, \varphi)}{r^2}.$$

The best weak direction maximizes  $g(\varphi)$ .

## The statistical comparison

D-optimal design maximizes joint Fisher information about ability and attention. In this example its weak-signal criterion is proportional to  $(4\beta - 1.5\alpha)^2$ , giving  $-69.4^\circ$ . The economic criterion values the **cost saved by changing incentives**.

# Interpreting the direction comparison

We compute the value of the **same signal** in two problems.

## Ability known

- The insurer tailors payments to ability and encourages hidden attention.
- A signal about ability alone has no value here.
- Here gain per  $r^2$  is approximately  $1.23 \cos^2 \varphi$  at weak precision.

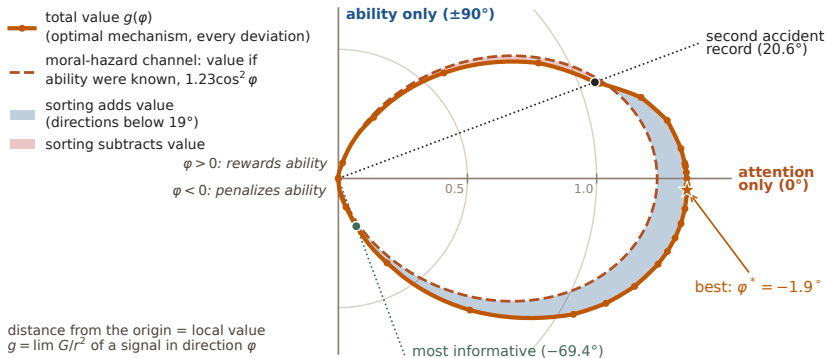
## Ability private

- The insurer also needs the driver to select the intended contract.
- The signal can change both attention incentives and contract choice.
- The difference in the two gains shows how private ability changes the signal's value.

## Reading the next figure

The dashed curve is value with ability known. The solid curve is value with private ability. Their gap can be positive or negative. Each problem compares with its own record-only benchmark.

# The best weak direction in this calibration



- **Angle is the measured mix. Radius is value:** savings  $\approx g(\varphi)r^2$  at a fixed small precision.
- Best:  $-1.9^\circ$ , almost pure attention. The D-optimal direction delivers 14.6% of this gain. Private ability slightly shifts the optimum from attention only.

## How much does the optimal signal save?

For weak monitoring in this calibration,

$$\varphi^* \approx -1.9^\circ, \quad G^*(r) \approx 1.351 r^2.$$

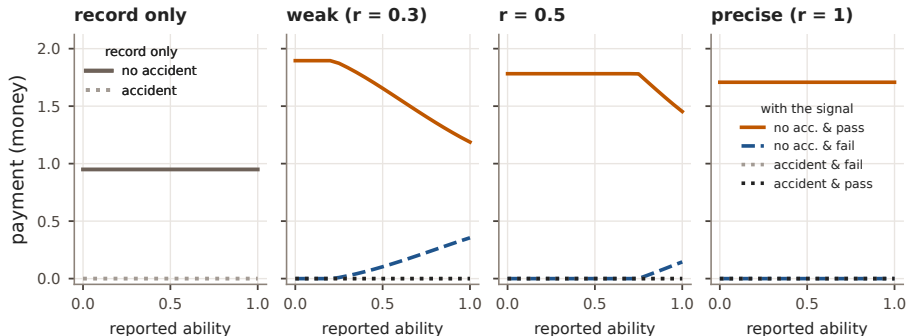
| precision $r$ | best direction | insurer saving | cost reduction |
|---------------|----------------|----------------|----------------|
| 0.05          | $-1.9^\circ$   | 0.00338        | 0.10%          |
| 0.3           | $-1.5^\circ$   | 0.133          | 3.9%           |
| 1             | $0.9^\circ$    | 0.788          | 23.1%          |

- Near zero precision, doubling  $r$  approximately quadruples the gain.
- At larger precision, we solve the problem again. Both the direction and the optimal payments can change.

Savings are in money per driver. The benchmark cost is 3.410. The quadratic expression is a weak-signal approximation. Table entries come from the full numerical problem.

# Optimal payments in the example

The benchmark pays 0.95 after no accident. With the best signal at each precision:



- **Weak signal: a menu.** Lower ability reports receive a larger pass/fail payment gap. Higher reports receive flatter contracts.
- **Precision 1: one contract.** Pay 1.71 after no accident and a pass, and zero otherwise.

All transfers are in money. The example pays zero after an accident at every reported precision.

## How a pass strengthens attention incentives

Fix ability and the attention level to be induced. Compare two rewarded outcomes:

|                                    | record only  | record and signal            |
|------------------------------------|--------------|------------------------------|
| rewarded outcome                   | no accident  | no accident and pass         |
| probability                        | $1 - p$      | $(1 - p)q$                   |
| proportional response to attention | $\gamma_a p$ | $\gamma_a p + \alpha(1 - q)$ |

- With  $\alpha > 0$ , more attention raises the chance of **both** avoiding an accident and passing.
- When paying only on the combined outcome is feasible, this stronger response reduces the expected utility payment needed for a given marginal incentive.
- A pure ability signal ( $\alpha = 0$ ) leaves this direct attention response unchanged. With private ability, it can still affect contract choice.

Proportional response means the derivative of log probability with respect to attention. Money costs also reflect payment risk. A weak signal generally rewards both pass and fail, as the previous slide shows.

## Why a small negative weight on ability helps here

The best small signal is almost pure attention, tilted slightly against ability:  $\varphi^* = -1.9^\circ$ .

1. **A tempting deviation.** An abler driver claims slightly less ability (a steeper contract) and keeps his own, lower attention level. Here this joint deviation binds.
2. **An attention signal deters it.** With his low attention level he passes less often, so the steep pass/fail pay is worth less to him.
3. **A negative ability weight helps further:** abler drivers pass less often at any attention level, and attention sensitivity falls only at second order ( $\cos^2 \varphi \approx 1 - \varphi^2$ ).

In general: what information needs, times what incentives demand

Under Theorem 2's conditions the best small signal is the top eigenvector of

$$M = \mathbb{E} \left[ \frac{\eta \eta^\top}{2 \bar{h}''} \right], \quad \eta(\theta) = (\mathcal{I}^y)^{-1} \Delta(\theta).$$

$(\mathcal{I}^y)^{-1}$ , the record's inverse Fisher information, is **what information needs**;  $\Delta = (c_a, U' + c_\theta)$ , the attention and screening requirements valued at the marginal cost of pay, is **what incentives demand**.

# How the new contracts change attention

In the benchmark, a common bonus induces  $a^0(\theta)$ . After a signal is added, each driver again chooses attention under the contract he selects.

## More effective attention incentives

- Payment can depend more closely on attention.
- With less costly incentives, the insurer may want to induce a higher level of attention.

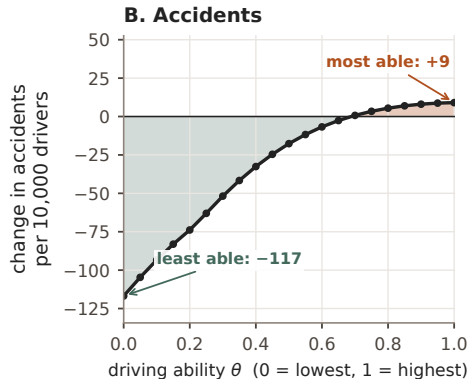
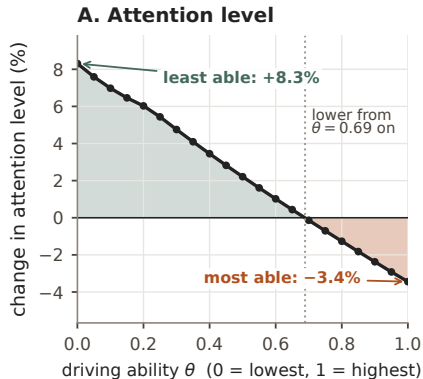
## Different incentives across abilities

- A menu can offer different bonus schedules to different drivers.
- Here the ability-known reference calls for stronger incentives at low ability and weaker incentives at high ability than the benchmark common bonus.

## The allocation question

How do these effects combine? We compare each driver's new choice with his **record-only attention level**.

# Attention under a weak signal



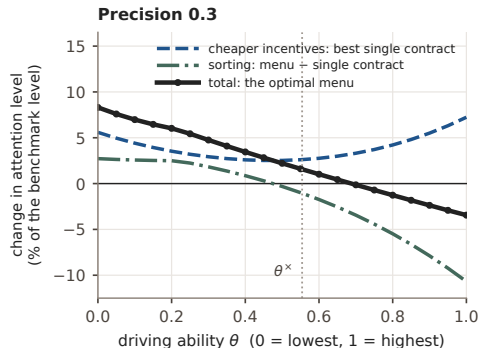
- At precision 0.3, attention rises by 8.3% for the least able and falls by 3.4% for the most able.
- Mean risk falls from 10.7% to 10.4%. Accident risk rises for some able drivers because their attention falls.

Best signal direction in this calibration. Changes are relative to the record-only benchmark.

# What allowing a menu changes

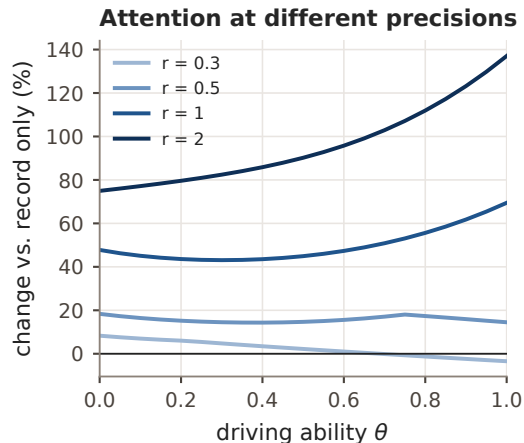
In this calibration, hold the **same weak signal** fixed and compare:

- **One contract for everyone.** The best common contract raises every driver's attention relative to the benchmark.
- **A menu of contracts.** Self-selection allows stronger incentives for the less able and flatter incentives for the abler.
- At high ability, this menu effect outweighs the increase under the common contract. Attention then falls below the benchmark.



Precision 0.3. Green: menu minus best common contract. Black: total change from the benchmark. The dotted line marks where ability-known and benchmark attention coincide.

# Attention as the signal becomes more precise



- In this calibration, a weak signal raises attention at low ability and lowers it at high ability.
- At higher precision, more effective incentives dominate in this example. At precision 1, every driver chooses more attention than in the benchmark.
- Higher-ability drivers continue to choose lower attention than lower-ability drivers, even when all choose more than in the benchmark.

Percent change from each driver's record-only attention level. The best direction is chosen separately at each precision.

## Precision 1: more attention at similar spending

In this calibration, the numerical solution returns one common contract. Compare it with the benchmark:

|                         | record only | with the signal      |
|-------------------------|-------------|----------------------|
| rewarded outcome        | no accident | no accident and pass |
| payment when rewarded   | 0.95        | 1.71                 |
| share receiving payment | 89%         | 49%                  |
| mean payment            | 0.85        | 0.84                 |
| mean accident risk      | 10.7%       | 7.4%                 |

- Approximately the same expected payment induces **47% more attention on average**. The reward now responds to attention through both the record and the signal.
- Payment is larger when it arrives and reaches fewer drivers. The driver bears more payment risk, which matters for welfare.

Payments are in money. Zero payment after all other outcomes. The next comparison considers the insurer, drivers, and total surplus.

# Who gains: rents, behavior, and payment risk

Monitoring changes three parts of the insurer's cost:

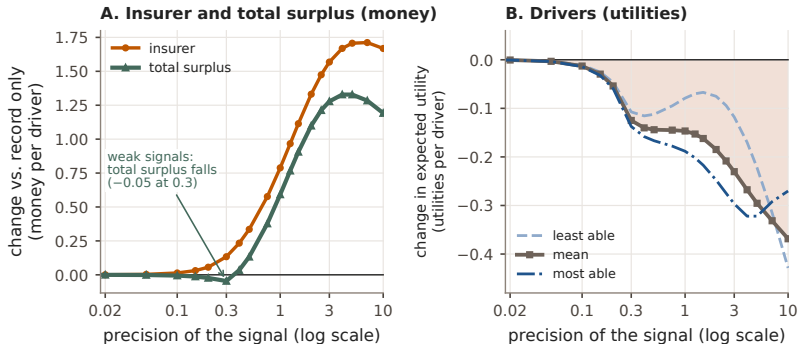
- **Driver rents.** Better targeted rewards can reduce extra driver utility. This benefits the insurer and can lower driver welfare.
- **Attention and accident losses.** More attention is costly but prevents accidents. Its money cost is the minimum certain compensation  $h(c)$ .
- **Payment risk.** Concentrating rewards on fewer outcomes can increase the cost of uncertain pay for risk-averse drivers.

## The welfare comparison

Total surplus rises when the fall in attention cost plus accident losses exceeds the extra payment risk cost. Rent extraction alone redistributes gains.

Driver welfare is expected utility. Surplus is in money and includes payment risk. The insurer can ignore the signal. Exact accounting is in the backup.

# Who gains in this calibration?



- **Weak monitoring ( $r = 0.3$ ):** insurer +0.13, total surplus -0.05. Better attention saves 0.07, but extra payment risk costs 0.12.
- **Higher precision ( $r = 1$ ):** insurer +0.79, total surplus +0.59. The larger attention improvement outweighs the payment risk cost.

Money per driver. Mean driver utility: -0.125 at  $r = 0.3$ , -0.147 at  $r = 1$ . All driver types lose utility in the computed solutions.

## Why a small negative weight on ability helps here

The best small signal is almost pure attention, tilted slightly against ability:  $\varphi^* = -1.9^\circ$ .

1. **A tempting deviation.** An abler driver claims slightly less ability (a steeper contract) and keeps his own, lower attention level. Here this joint deviation binds.
2. **An attention signal deters it.** With his low attention level he passes less often, so the steep pass/fail pay is worth less to him.
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$$M = \mathbb{E} \left[ \frac{\eta \eta^\top}{2 \bar{h}''} \right], \quad \eta(\theta) = (\mathcal{I}^y)^{-1} \Delta(\theta).$$

$(\mathcal{I}^y)^{-1}$ , the record's inverse Fisher information, is **what information needs**;  $\Delta = (c_a, U' + c_\theta)$ , the attention and screening requirements valued at the marginal cost of pay, is **what incentives demand**. Here the record is one index ( $\mathcal{I}^y$  is singular), so the tilt comes from the binding joint deviation.

# Takeaways

## 1. What to measure depends on the incentive problem.

In this calibration, almost pure attention is best for weak monitoring. The D-optimal direction delivers only 14.6% of its value.

## 2. Better monitoring changes who supplies attention.

Here weak monitoring raises attention for the less able and lowers it for some abler drivers. At precision 1, attention rises for everyone.

## 3. Welfare depends on behavior and payment risk.

Here driver utility falls. Total surplus falls with weak monitoring and rises when the attention gains become larger.

### The general lesson

The value of additional information depends on the existing information and the economic incentives it changes.

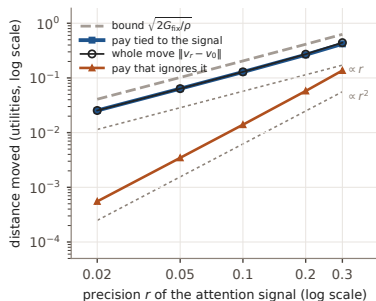
# Step 1: why payments stay close

Fix benchmark attention  $a^0$ . Let  $v^0$  be the old contract,  $v^r$  its cheapest full-IC replacement, and  $G_{\text{fix}}$  the saving.

1. **Ignoring the signal is feasible.** The old payments preserve every report–attention payoff.
2. **Changing payments has a quadratic cost.** IC is linear in utilities at fixed attention, and  $h'' \geq \rho > 0$ .
3. **The optimal signal-linked payment is small.** A payment gap  $w$  gains order  $rw$  in incentives and costs order  $w^2$  in risk. Balancing them gives  $w = O(r)$  and savings  $O(r^2)$ .

$$\frac{\rho}{2} \|v^r - v^0\|^2 \leq G_{\text{fix}} = O(r^2).$$

Thus the payment change is  $O(r)$ .



The  $O(r^2)$  gain requires the local expansion or a suitable constraint qualification. In the example, signal-dependent pay moves by  $O(r)$  and average pay by  $O(r^2)$ .

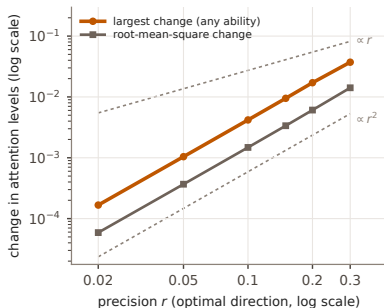
## Step 2: why attention stays close

The allocation is the attention assigned to each ability.  
Stability is an **assumption about the benchmark**,  
beyond convexity in payments.

1. Moving distance  $d$  from the benchmark raises record-only cost by at least  $\kappa d^2$ .
2. A weak signal saves at most  $Cr^2$  uniformly over the feasible allocations in the comparison.
3. Hence an optimal allocation satisfies  $\kappa d^2 \leq Cr^2$ , so  $d = O(r)$ .

With regular local geometry, changing attention adds only a higher-order correction:

$$G = G_{\text{fix}} + O(r^3).$$



The insurance example is outside the regular case. Its attention moves by  $O(r^2)$ , yet re-optimizing attention adds about 23% of weak-signal value near the best direction.

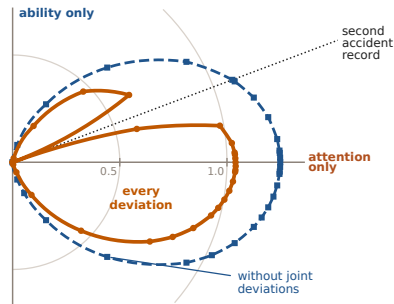
## Step 3: when off-path choices have a smaller effect

### The local argument

- Changing both report and attention adds a mixed term of order  $(\hat{\theta} - \theta)(\hat{a} - a(\hat{\theta}))$ .
- **Regular incentive geometry** controls this term. Under Theorem 1's conditions, screening and on-path obedience give the same second-order value as full IC.

### Why the example keeps full IC

- At the pooled benchmark, any report combined with one's own attention gives the same payoff.
- Dropping off-path choices overstates attention-signal value by about 20% at fixed attention.



Radius is gain divided by  $r^2$ , with attention fixed and  $r = 0.02$ . Full IC (solid) makes a second accident record almost valueless at this allocation. Re-optimizing attention gives it positive value.

## Step 4: the general direction criterion

Assume a stable benchmark, regular incentives, a record informative about both hidden variables, and slack limited liability. Then for a weak pass/fail signal:

$$G = \frac{1}{4}(\alpha, \beta)M(\alpha, \beta)^\top + o(r^2).$$

- $M$  weights signal sensitivity by the benchmark's costly incentive requirements and the cost of payment risk.
- Under  $\alpha^2 + \beta^2 \leq r^2$ , the best direction is the top eigenvector of  $M$ . With a weighted capacity constraint, it is a generalized eigenvector.
- The direction can favor ability, attention, or opposite signs on the two. **The matrix depends on the economic problem.**

### How the example differs

Its pooled benchmark, rank-one record, and binding limited liability fall outside these conditions. We maximize its full-IC numerical value directly.

## Backup: Theorems 1 and 2, full statements

### Theorem 1 (at the benchmark allocation)

Assume (R1) the benchmark contract  $v_0$  has regular incentive geometry, or (R2) expected likelihood ratios are linear in  $(a, \theta)$  and  $v_0$  is IC. Then for small  $r$ : the benchmark contract re-priced for the signal,  $\tilde{v}_r$ , is fully IC;  $\tilde{G} \leq G_{\text{fix}}^F \leq G_{\text{fix}}^{\text{SE}} \leq G_{\text{fix}}^{\text{FOC}} \leq \tilde{G} + \kappa_U r^4$  ( $\kappa_U$  a constant;  $\tilde{G}$ : the saving of  $\tilde{v}_r$ ); and near-optimal F, SE and FOC contracts are within  $O(r^2)$  of  $\tilde{v}_r$ .

### Theorem 2 (optimal marginal monitoring)

Assume benchmark stability,  $\mathcal{I}^y \succeq \iota I$ , slack limited liability and Theorem 1's conditions;  $M$  is defined on the next slide. Let  $\nu_1 = \max\{z^\top M z : z^\top \Omega z = 1\}$ , attained at  $z_1$ . The best value of a signal of precision  $r$  is  $\mathcal{I}_s(0) \nu_1 r^2 + o(r^2)$ ; if  $\nu_1 > 0$  optimal designs exhaust capacity, and if  $\nu_1$  is simple, the optimal design satisfies  $(\alpha, \beta)/r \rightarrow \pm z_1$ , with  $M z_1 = \nu_1 \Omega z_1$ .

- F: all deviations; SE: screening and on-path obedience; FOC: first-order conditions.  $G_{\text{fix}}^J$ : the money saved at the benchmark allocation under system  $J$ .
- Joint deviations: the slack of a joint deviation is the screening slack plus the attention-level slack minus a mixed second difference of order  $(\hat{\theta} - \theta)(\hat{a} - a(\hat{\theta}))$ .